

Ankush Mondal

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EDUCATION

Techno India University

Bachelor of Technology in Computer Science and Business System

Kolkata, India

Sep. 2022 – Aug 2026

EXPERIENCE

Open Source Contributions

TursoDB, PostgreSQL, FreeBSD

- Added **.read** command to execute *SQL* from a file in TursoDB. [Link](#)
- Migrated Turso core from custom SmallVec to **smallvec** crate. [Link](#)
- Implemented **PostgreSQL protocol logging** to debug dump failures. [Link](#)
- fixed POSIX block-size handling in FreeBSD **df -k** flag. [Link](#)
- update bash script to support **gcc compilation** without coverage. [Link](#)
- Fixed **unix domain socket** sharing issue causing high CPU consumption. [Link](#)

Cyber Security IAM Intern

EY GDS (Onsite)

Feb 2026 - May 2026

- Worked with Identity and Access Management (IAM) systems including **CyberArk**, **Azure AD**, and **SailPoint** for user access control and privilege management.
- Gained hands-on experience with **LDAP** for directory services, authentication, and user provisioning workflows.
- Assisted in managing secure access policies, onboarding/offboarding processes, and resolving access-related issues in enterprise environments.

Linux System Programming Trainee

Sasken Technologies Ltd (Remote)

June 2025 - Aug 2025

- Developed a monitoring tool in **C** for **Linux systems** to detect kernel memory leaks by parsing the `/proc/slabinfo` and `/proc/vmstat` virtual files.

PROJECTS

GhostSSH: SSH over HTTPS Tunnel | C, WebSockets, TCP, HTTPS, [Link](#)

- Built a lightweight tool to tunnel SSH traffic over **secure WebSocket (WSS)** connections, enabling access in restricted networks.
- Implemented a **bidirectional TCP-to-WebSocket bridge** to stream SSH data in real-time without modifying the SSH server.
- Designed a firewall-friendly solution running over **HTTPS (port 443)** that works seamlessly with standard SSH clients.

Zail: Realtime Filesystem & Journal Monitor | Zig, Linux Syscalls, Epoll, Inotify, [Link](#)

- Built a **low-latency event-driven monitor** using Linux `epoll`, `inotify`, and `signalfd` to track filesystem changes, system signals, and systemd journal updates in a unified loop.
- Engineered a **recursive directory-watch system** with dynamic watch-descriptor mapping and path caching for robust large-tree monitoring.

Vec.hpp | C++, Makefile, Memory Management, [Link](#)

- Built a custom `Vec<T>` class with a **chunked paging system** using low-level `mmap`, enabling dynamic expansion without costly reallocations.
- Designed scalable memory architecture with **O(1) element access** via bitwise operations, placement `new`, and pointer arithmetic, minimizing fragmentation.
- Benchmarked up to **500M elements**, achieving a **1.73× speedup** in insertion throughput over `std::vector`.

TECHNICAL SKILLS

Languages & tools: C/C++, Py, go, Rust, SQL, Linux, Docker, GDB, Makefiles, Bash

Courses: OS, DBMS, TCP/IP, Distributed Systems, Theory of Computation

Skills: Systems Programming, Memory Management, Data Structures and algorithms, Performance Optimization

Coding Profile (Link): More than 300 problems solved including Hashing, tree, graph, Linkedlist, DP